

Date Functions

Table 1: Employees

emp_id	name	hire date
1	Alice	2020-01-15
2	Bob	2021-06-10
3	Charlie	2023-03-22

Q1 Add 6 months to each employees hire date using DATEADD
Expected columns: emp_id, name, hire_date, hire plus 6 months

```
SELECT emp_id,  
       name,  
       hire_date,
```

```
DATEADD(Month, 6, hire_date) AS hire plus 6 months
```

```
FROM Employees;
```

EmpId	name	hire date	hire plus 6 months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Q2 Ages in days from dob to today

```
SELECT student_id,  
       name
```

```
DATEDIFF(DAY, dob, CURRENT_DATE) AS age in days
```

```
FROM Students;
```

Output (Assuming today = 2025-10-13)

Student_id	name	age in days
101	Maya	7375
102	Ethan	7669
103	Sienna	7169

Q3:

Events

Event_id	event_name	event_date
1	Seminar	2024-06-15
2	Workshop	2025-07-10
3	Hackathon	2025-07-20

Find how many days are left until each event using DATEDIFF()

Event_id	event_name	day remaining
1	Seminar	-485
2	Workshop	-42
3	Hackathon	-266

SELECT event_id,

event_name

DATEDIFF(Day, Current_date, event_date) As day-remaining

FROM Events;

Q4 Invoices

Invoice_id	Issue_date	due date
501	2025-03-10	2025-03-25
502	2025-04-01	2025-04-15
503	2025-04-10	2025-04-20

Calculate the number of days between issuedate and due date

SELECT invoice_id,

issue_date,

due date,

DATEDIFF(Day, issue_date, due date)

FROM Invoices

Invoice_id	Issuedate	due date	days-between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Q5 Courses

Course_id	name	startdate
201	SQL Basics	2025-05-01
202	Python	2025-06-01

Format start date as Month YYYY using TO_CHAR()

```
SELECT course_id,  
       name,  
       TO_CHAR(start_date, 'Month YYYY')  
FROM   courses;
```

Course_id	name	Formatteddate
201	SQL Basics	May 2025
202	Python	June 2025

Q6 Memberships

Member_id	Start year	start month	start_day
1	2023	5	10
2	2022	11	25

Create full date from parts using DATE FROM PARTS

```
SELECT member_id  
       DATEFROMPARTS(start_year, start_month, start_day)  
       AS full_start_date  
FROM   Memberships;
```

Member_id	Full start_date
1	2023-05-10
2	2022-11-25

Q7: Subscriptions

Sub-id	Plan	Renewal date
11	Basic	2025-01-01
12	Premium	2025-03-15

Extend each renewal date by 1 year using DATEADD()

```
SELECT sub-id,  
       Plan,  
       DATEADD(YEAR, 1, renewal-date) AS Extended renewal date  
FROM Subscriptions;
```

Sub-id	Plan	Extended renewal date
11	Basic	2026-01-01
12	Premium	2026-03-15

Q8 Orders

Order-id	order-date
1001	2025-04-15
1002	2025-04-10

Show current date & difference from order-date
Use current_date and DATEDIFF()

```
SELECT order-id,  
       order-date,
```

Current DATE AS Today date,

```
DATEDIFF(DAY, order-date, CURRENT DATE) AS day since  
order
```

```
FROM order;
```

order_id	order date	today date	Days since order
1001	2025-04-15	2025-10-13	181
1002	2025-04-10	2025-10-13	186

Q9

```

SELECT training_id,
       topic
       EXTRACT (YEAR FROM training date) AS training_year
FROM Trainings;

```

Training-id	Topic	Training year
1	Safety	2025
2	Compliance	2025

Q10

```

SELECT post_id,
       title,
       EXTRACT (HOUR FROM published on) AS hour published
       EXTRACT (MINUTE FROM published on) AS minute
       published
FROM Blog-Posts;

```

Post-id	title	hour published	minutes-published:
1	SQL Tips	10	15
2	Data cleaning	16	45

Q11

```
SELECT driver_id,
       license_expiry,
       DATEDIFF (DAY, CURRENT DATE, license_expiry) AS day_left
FROM Drivers;
```

Driver_id	License_expiry	Day left
301	2025-08-10	-64
302	2023-12-31	-652

Q12

```
SELECT message_id,
       sent_timestamp,
       CURRENT_TIMESTAMP AS current_timestamp,
       DATEDIFF (Second, sent_timestamp, CURRENT_TIMESTAMP)
       AS seconds_since_sent
FROM Messages;
```

Message_id	Sent_timestamp	Current timestamp
1	2025-04-19 09:32:45	2025-10-13 13:00:00
2	2025-04-18 23:59:59	2025-10-13 13:00:00

Q13

```
SELECT return_id,
       return_date,
       DATEADD (DAY, 15, return_date) AS restock_date
FROM Returns;
```

return_id	return_date	Restock date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Q14

SELECT Assign-id,
 TO DATE (Assigned on, 'YYYY-MM-DD') AS
 assigned on date
 FROM Assignments;

Assign id	Assigned on-date
1	2025-03-01
2	2025-03-05

Q15

SELECT meeting_id,
 TO CHAR (scheduled-time, 'Month DD, YYYY at
 HH:MM AM') AS formatted_meeting_time
 FROM meetings;

meeting id	Formatted meeting time
1	April 19, 2025 at 02:00 PM
2	April 19, 2025 at 09:30 AM

1 Show all employees with (0 if NULL)

```
SELECT employee_id,  
       name,  
       IFNULL (Salary, 0) AS salary with default  
FROM   employees;
```

Employee-id	Name	Salary with default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

2 Department name or "Not Assigned"

```
SELECT employee_id,  
       name,  
       IFNULL (Department, 'Not Assigned') AS Department  
FROM   employees;
```

1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

3. Orders with NULL customer id

```
SELECT order_id, customer_id  
FROM   Orders  
WHERE  customer_id IS NULL;
```

Order id	Customer_id
103	NULL

4 Delivery status(Pending if NULL)

```
SELECT order_id  
       customer_id  
       IFNULL(TO_CHAR(delivery_date, 'YYYY-MM-DD'), Pending)  
       AS delivery_status  
FROM Orders;
```

Order id	Customer id	Delivery status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

5. Replace NULL grades with 0

```
SELECT student_id,  
       name,  
       IFNULL(Grade, 0) AS final_grade  
FROM Students;
```

Student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

6. Count students with NULL grade

```
SELECT count(*) AS not_graded_count  
FROM students  
WHERE grade IS NULL
```

Not graded count
1

Table 4: Products

7. Final price after discount

```
SELECT product_id,  
       name,  
       Price - IFNULL (discount, 0) AS final price  
FROM Products;
```

Product_id	name	final price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Table 5

8. Count customers with no email

```
SELECT count(*) AS missing_email_count  
FROM Customers
```

WHERE email IS NULL;

Missing_email_count
2

9. Show all customers with email or NO email

```
SELECT customer_id,  
       name
```

```
IF NULL (email, 'No email') AS email display  
FROM Customers;
```

Customer_id	name	email display
1	Linda	No email
2	Joseph	Joseph@email.com
3	Nia	No email

Table 6

10. Replace NULL method with Unknown

```
SELECT payment_id,
       IFNULL(Method 'Unknown') AS Method display,
       status
FROM payments;
```

Payment_id	Method display	Status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed

Table 7: Inventory

Item_id	Item name	Quantity
1	Pen	NULL
2	Notebook	150
3	Eraser	NULL

```
SELECT item_id,
       item_name,
       IFNULL(Quantity, 0) AS quantity checked
```

FROM Inventory;		
1	Pen	0
2	Notebook	150
3	Eraser	0

Table 8 Employee-Extra

12.

OR commission

SELECT emp-id,

COALESCE(bonus, commission, 0) AS first available reward

FROM Employees_Extra;

Emp-id	First available reward
1	300
2	100
3	0

Table 9: Classes

Q13

SELECT count(*) AS no room count

FROM Classes

WHERE room IS NULL;

no-room count

2

Table 10: Attendance

Q14

SELECT student-id,
date,

IFNULL(status, 'Not Marked') AS Attendance
status

FROM Attendance

Student id	date	Attendance status
1	2025-04-01	Not marked
2	2025-04-01	Present
3	2025-04-01	Absent

Table 11 : Bank Accounts

Q15

```
SELECT account_id,
       IFNULL (Account_type 'Unknown') AS type_display,
       IFNULL (Balance, 0) AS balance_checked
FROM Bank accounts;
```

Account_id	type display	balance checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Table 12 Projects

Q16

```
SELECT project_id,
       title,
       IFNULL (TO CHAR (Start_date 'YYYY-MM-DD') TBD
       AS start display
FROM projects;
```

Projects_id	title	Start display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Q17

```
SELECT review-id,
       Product-id,
       IF NULL (comment, 'No comment') AS comment_display,
       IF NULL (rating, 0) AS rating_display
```

FROM Reviews;

Review-id	Product-id	Comment-display	rating-display
1	501	Great product	4
2	502	No comment	0
3	503	Work fine	3

Q18

```
SELECT supplier-id,
       name,
```

COALESCE (Phone, Alt-phone, No contact) AS contact number

FROM Suppliers;

Supplier-id	Name	Contact number
1	Global goods	123456789
2	Best supplies	987654321
3	Value source	No contact

Q19

```
SELECT user-id,
```

IF NULL (theme 'Light') AS timeset,

IF NULL (Language 'English') AS Language set,

IF NULL (timezone UTC) AS timezone set

FROM User settings

Q17

```
SELECT review_id,
       Product_id,
       IF NULL (comment, 'No comment') AS comment_display,
       IF NULL (rating, 0) AS rating_display
```

Review_id	Product_id	comment_display	rating_display
1	501	Great product	4
2	502	No comment	0
3	503	Work fine	3

Q18

```
SELECT supplier_id,
       name,
       COALESCE (Phone, Alt_phone, No contact) AS contact number
FROM Suppliers;
```

Supplier_id	Name	Contact number
1	Global goods	123456789
2	Best supplies	987654321
3	Value source	No contact

Q19

```
SELECT user_id,
       IF NULL (theme, 'Light') AS theme_set,
       IF NULL (Language, 'English') AS Language set,
       IF NULL (timezone, 'UTC') AS timezone set
FROM User settings
```

User-id	Theme set	Language set	time zone
1	Light	English	UTC
2	Dark	English	UTC +1
3	Light	English	UTC

Q 20

```

SELECT record id,
       machine-id,
       IFNULL (issue, unknown issue) AS issue log,
       IFNULL (technical 'Not Assigned') AS Technical name
FROM Maintenance;

```

record id	Machine id	issue log	Technical name
1	M101	Overheating	Not assigned
2	M102	Unknown issue	Not assigned
3	M103	Jammed	Alex